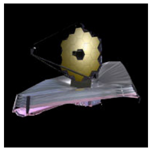

7230 - Shadows and Dust - MIREX Mapping of a Massive Molecular Filament

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
G35				
	1	NIRCAM 1 field	NIRCam Imaging	(3) G35
	4	NIRCAM 1 field	NIRCam Imaging	(3) G35
	2	MIRI mosaic	MIRI Imaging	(3) G35
	3	MIRI mosaic - BKG	MIRI Imaging	(4) G35-BKG

ABSTRACT

We propose MIRI and NIRCam imaging of a well-studied Infrared Dark Cloud (IRDC) filament that is one of the best test cases for mid infrared extinction (MIREX) mapping. This project will produce MIREX maps at <0.3 arcsecond resolution, eight times finer than achieved with Spitzer and a unique and powerful probe of the structure of star-forming clouds that only JWST can deliver. The NIR images will detect background stars, whose sight lines will yield pinpoint extinction measurements that help calibrate the MIREX map. Combined analysis of NIR and MIR extinctions will test models of dust grain evolution in a wide range of conditions in molecular gas. The same NIR and MIR images will also be exceptionally sensitive to protostellar and young stellar object (YSO) populations in the cloud, enabling a full characterization of the initial mass function (IMF) in this early-stage massive molecular filament. Key questions that will be answered by this project include:

- What are the structural properties of molecular gas at the onset of massive star and cluster formation?

- What are the natures of the young and forming stellar populations in this filamentary IRDC?

- What is the core mass function (CMF) in this region and how does it relate to the stellar initial mass function (IMF)?

- How does the NIR to MIR extinction law vary as a function of density and temperature and what are the implications for dust grain growth and evolution?

OBSERVING DESCRIPTION

This program will carry out NIRCam (F115W, F162M, F182M, F200W, F356W, F405N, F470N, F480M) and MIRI (F770W and F2100W) imaging of a filamentary Infrared Dark Cloud (IRDC) to carry out MIR extinction (MIREX) mapping, including NIR extinction calibration, and characterize its protostellar and YSO populations. Coordinated parallels are requested to improve modeling of the background diffuse MIR emission that is crucial for MIREX mapping and to characterize the field star population in these wavebands, which is important for NIR extinction methods in the Galactic plane and for estimating on-source levels of field star contamination.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	G35	RA: 18 57 8.0000 (284.2833333d) Dec: +02 10 10.00 (2.16944d) Equinox: J2000		
Comments: *Category=ISM* *Description=[Dense interstellar clouds]*				
(4)	G35-BKG	RA: 18 58 31.0000 (284.6291667d) Dec: +02 21 36.00 (2.36000d) Equinox: J2000		
Comments: *Category=ISM* *Description=[Dense interstellar clouds]*				

Observation

Proposal 7230, Observation 1: NIRCAM 1 field

Fri Apr 24 20:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Coordinated Parallel Template(s): MIRI Imaging

Diagnostics

(NIRCAM 1 field (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	G35	RA: 18 57 8.0000 (284.2833333d) Dec: +02 10 10.00 (2.16944d) Equinox: J2000		

Comments:
Category=ISM
Description=[Dense interstellar clouds]

Template

NIRCam Imaging	MIRI Imaging
Module: ALL	Subarray: FULL
Subarray: FULL	
Target Placement: Module gap (large extended source)	

Dithers

#	Primary Dither Type	Primary Dithers	Dither Size	Subpixel Positions	Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
1	FULLBOX	6		1	NIRCam Only	NO_DITHERING

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F162M+F150W2	F405N+F444W	SHALLOW2	6	1	6	6	1739.357	145645
2	F182M	F470N+F444W	SHALLOW2	6	1	6	6	1739.357	145645
3	F115W	F480M	SHALLOW2	6	1	6	6	1739.357	145645
4	F200W	F356W	BRIGHT2	6	1	6	6	773.047	145645

Spectral Elements

MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	10	1	1		6	6	166.502	
2	F770W	FASTR1	10	1	1		6	6	166.502	
3	F1500W	FASTR1	10	1	1		6	6	166.502	
4	F2100W	FASTR1	10	1	1		6	6	166.502	

Special Requirements	Aperture PA Range 249.92542306 to 269.92542306 Degrees (V3 250.0 to 270.0) No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible

Observation

Proposal 7230, Observation 4: NIRCAM 1 field

Fri Apr 24 20:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Coordinated Parallel Template(s): MIRI Imaging

Diagnostics

(NIRCAM 1 field (Obs 4)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	G35	RA: 18 57 8.0000 (284.2833333d) Dec: +02 10 10.00 (2.16944d) Equinox: J2000		

Comments:
Category=ISM
Description=[Dense interstellar clouds]

Template

NIRCam Imaging	MIRI Imaging
Module: ALL	Subarray: FULL
Subarray: FULL	
Target Placement: Module gap (large extended source)	

Dithers

#	Primary Dither Type	Primary Dithers	Dither Size	Subpixel Positions	Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
1	FULLBOX	6		1	NIRCam Only	NO_DITHERING

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F162M+F150W2	F405N+F444W	SHALLOW2	6	1	6	6	1739.357	145645
2	F182M	F470N+F444W	SHALLOW2	6	1	6	6	1739.357	145645
3	F115W	F480M	SHALLOW2	6	1	6	6	1739.357	145645
4	F200W	F356W	BRIGHT2	6	1	6	6	773.047	145645

Spectral Elements

MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	10	1	1		6	6	166.502	
2	F770W	FASTR1	10	1	1		6	6	166.502	
3	F1500W	FASTR1	10	1	1		6	6	166.502	
4	F2100W	FASTR1	10	1	1		6	6	166.502	

Special Requirements	
	Aperture PA Range 249.92542306 to 269.92542306 Degrees (V3 250.0 to 270.0) No Parallel Attachments

Observation

Proposal 7230, Observation 2: MIRI mosaic

Fri Apr 24 20:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Coordinated Parallel Template(s): NIRCam Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	G35	RA: 18 57 8.0000 (284.2833333d) Dec: +02 10 10.00 (2.16944d) Equinox: J2000		

Comments:
Category=ISM
Description=[Dense interstellar clouds]

Template

MIRI Imaging	NIRCam Imaging
Subarray: FULL	Module: B Subarray: FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
1	4	10.0	0.0	-1.429	0.0	DEFAULT

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT

Spectral Elements

MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F770W	FASTR1	10	2	1	Dither 1	4	8	233.103	145666
2	F2100W	FASTR1	10	2	1	Dither 1	4	8	233.103	145666

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F470N+F444W	BRIGHT2	2	1	4	4	171.788	
2	F200W	F356W	BRIGHT2	2	1	4	4	171.788	

Special Requirements	
	Group Visits within 53.0 Days Aperture PA Range 264.83544897 to 99.83544897 Degrees (V3 260.0 to 95.0) Visits Same PA No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible

Observation

Proposal 7230, Observation 3: MIRI mosaic - BKG

Fri Apr 24 20:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Coordinated Parallel Template(s): NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	G35-BKG	RA: 18 58 31.0000 (284.6291667d) Dec: +02 21 36.00 (2.36000d) Equinox: J2000		

Comments:
Category=ISM
Description=[Dense interstellar clouds]

Template

MIRI Imaging	NIRCam Imaging
Subarray: FULL	Module: B Subarray: FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT

Spectral Elements

MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F770W	FASTR1	10	2	1	Dither 1	4	8	233.103	
2	F2100W	FASTR1	10	2	1	Dither 1	4	8	233.103	

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F470N+F444W	BRIGHT2	2	1	4	4	171.788	
2	F200W	F356W	BRIGHT2	2	1	4	4	171.788	

Special Requirements	
	No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible